OUTLINE

- Simple linear regression: Descriptive
  - How strong is the linear relationship? (Correlation)
  - Finding the line of best fit
  - How good is the fit?
- Simple linear regression: Inferential
  - Assumptions

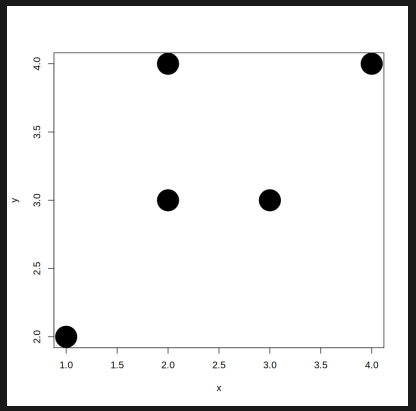
SIMPLE LINEAR REGRESSION: DESCRIPTIVE

- How strong is the linear relationship? (Correlation)
- Finding the line of best fit
- How good is the fit?

HOW STRONG IS THE LINEAR RELATIONSHIP? (CORRELATION)

Problem. Draw scatterplot of the following points. Then find and interpret the strength of the linear relationship:

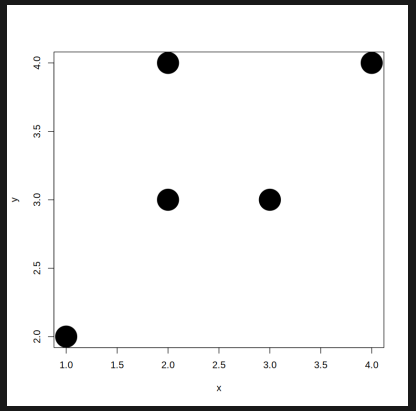
x	**1**	**2**	**2**	**3**	**4**
y	2	3	4	3	4



Answer: Scatterplot is drawn with PROC PLOT

```
data points;
    infile datalines dsd;
    input x y;
datalines;
1, 2
2, 3
2, 4
3, 3
4, 4
;

proc sgplot data=points;
    scatter x=x y=y;
run;
```



Answer: Find correlation with PROC CORR

```
proc corr data=points;
run;
```

The CORR Procedure

2 Variables:	x y

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
x	5	2.40000	1.14018	12.00000	1.00000	4.00000
y	5	3.20000	0.83666	16.00000	2.00000	4.00000

Pearson Correlation Coefficients, N = 5
Prob > |r| under H0: Rho=0

	x	y
x	1.00000	0.68139 0.2053
y	0.68139 0.2053	1.00000

Interpretation: Since correlation $r = 0.68$, there is strong (linear) association between x and y.

Interpreting correlation coefficient values [edit]

The correlation between two variables have different associations that are measured in values such as r or R. Correlation values range from −1 to +1, where ±1 indicates the strongest possible correlation and 0 indicates no correlation between variables.[5]

r or R	r or R	Strength or weakness of association between variables[6]
+0.8 to +1.0	-1.0 to -0.8	Perfect or very strong association
+0.6 to +0.8	-0.8 to -0.6	Strong association
+0.4 to +0.6	-0.6 to -0.4	Moderate association
+0.2 to +0.4	-0.4 to -0.2	Weak association
+0.0 to 0.2	-0.2 to 0.0	Very weak or no association

https://en.wikipedia.org/wiki/Correlation_coefficient

Correlation $r = \cos(\theta_{x,y})$ measures the direction and strength of linear relationship:

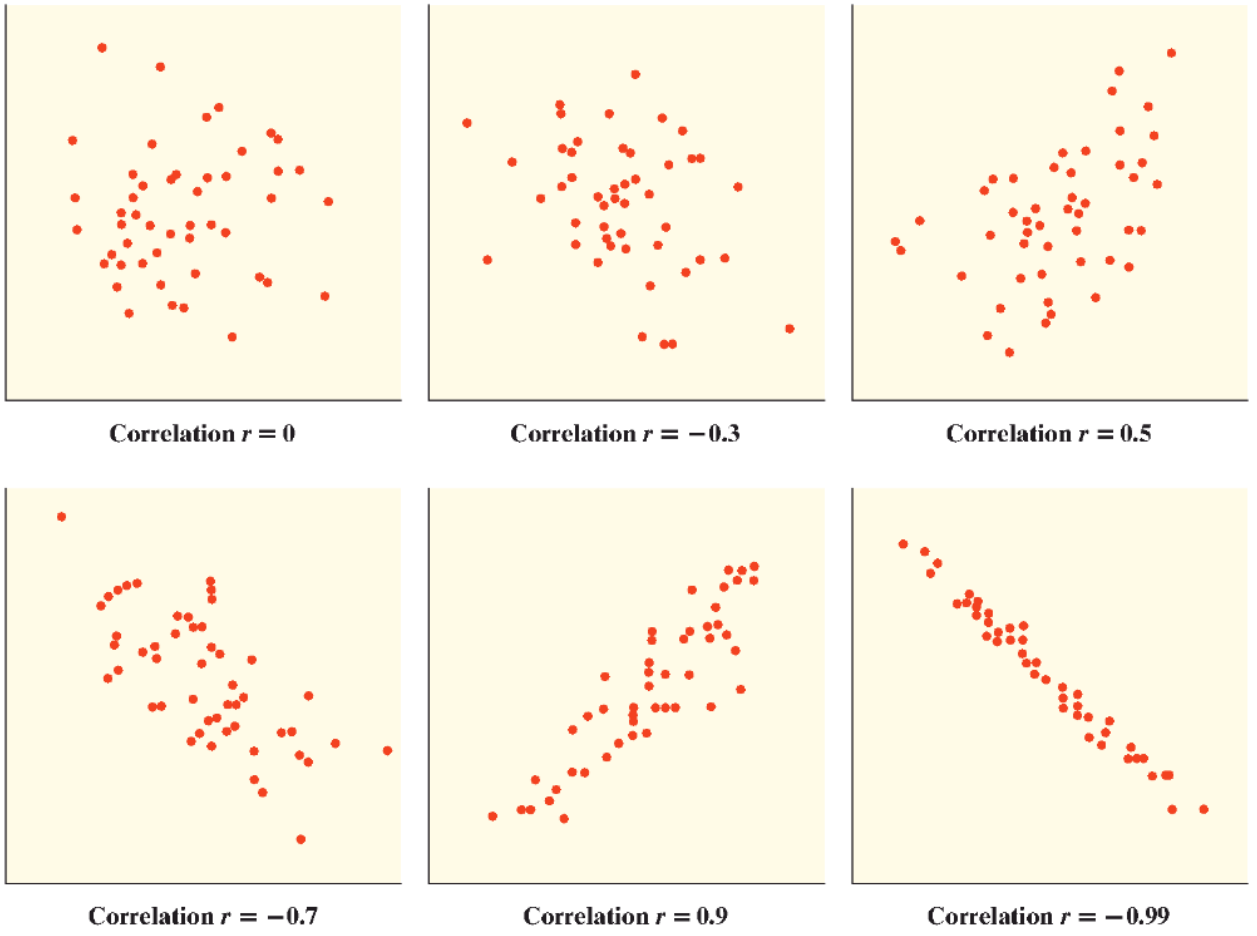
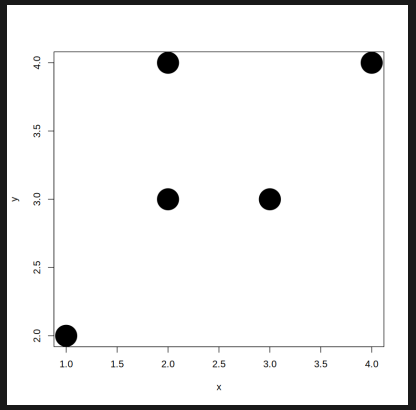


FIGURE 3.6 How correlation measures the strength of a linear relationship. Patterns closer to a straight line have correlations closer to 1 or -1.

FINDING THE LINE OF BEST FIT: THEORY

Problem. Find the best fitting line (in the least squares sense) that pass through the following points:

x	1	2	2	3	4
y	2	3	4	3	4



Problem. Find the best fitting line (in the least squares sense) that pass through the following points:

x	**1**	**2**	**2**	**3**	**4**
y	2	3	4	3	4

Model:

$$y = ax + b$$

x	**1**	**2**	**2**	**3**	**4**
y	2	3	4	3	4

Fitted model (with additive error/residual term):

$$
\begin{aligned}
2 &= a \cdot 1 + b + e_1 \\
3 &= a \cdot 2 + b + e_2 \\
4 &= a \cdot 2 + b + e_3 \\
3 &= a \cdot 3 + b + e_4 \\
4 &= a \cdot 4 + b + e_5
\end{aligned}
$$

Fitted model as matrices:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Fitted model as matrices:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix}}_{\begin{array}{c}\text{design} \\ \text{matrix } X\end{array}} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Fitted model as matrices:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Choice of $\vec{a} = (a, b)$ that minimizes Euclidean length of error vector:

$$\vec{y} = X\vec{a} \Rightarrow X^T\vec{y} = X^T X\vec{a}$$

$$\Rightarrow \boxed{(X^T X)^{-1}(X^T\vec{y}) = \vec{a}}$$

$$\vec{y} = X\vec{a} \Rightarrow X^T\vec{y} = X^TX\vec{a}$$

$$\Rightarrow \boxed{(X^TX)^{-1}(X^T\vec{y}) = \vec{a}}$$

$$\Rightarrow a = 0.5,\ b = 1$$

$$\vec{y} = X\vec{a} \Rightarrow X^T\vec{y} = X^TX\vec{a}$$

$$\Rightarrow \boxed{(X^TX)^{-1}(X^T\vec{y}) = \vec{a}}$$

$$\Rightarrow a = 0.5,\ b = 1$$

Our model is $y = 0.5x + 1$.

Our fitted model is:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Our fitted model is:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Our fitted model is:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \times 0.5 + 1 \times 2 \\ 2 \times 0.5 + 1 \times 2 \\ 2 \times 0.5 + 1 \times 2 \\ 3 \times 0.5 + 1 \times 2 \\ 4 \times 0.5 + 1 \times 2 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Our fitted model is:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 3 \\ 3 \\ 3.5 \\ 4 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \end{bmatrix}$$

Our fitted model is:

$$\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 3 \\ 3 \\ 3.5 \\ 4 \end{bmatrix} + \begin{bmatrix} -0.5 \\ 0 \\ 1 \\ -0.5 \\ 0 \end{bmatrix}$$

Our fitted model is:

$$\underbrace{\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix}}_{y} = \underbrace{\begin{bmatrix} 2.5 \\ 3 \\ 3 \\ 3.5 \\ 4 \end{bmatrix}}_{\hat{y}} + \underbrace{\begin{bmatrix} -0.5 \\ 0 \\ 1 \\ -0.5 \\ 0 \end{bmatrix}}_{\text{residual vector}}$$

$$\underbrace{\begin{bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 4 \end{bmatrix}}_{y} = \underbrace{\begin{bmatrix} 2.5 \\ 3 \\ 3 \\ 3.5 \\ 4 \end{bmatrix}}_{\hat{y}} + \underbrace{\begin{bmatrix} -0.5 \\ 0 \\ 1 \\ -0.5 \\ 0 \end{bmatrix}}_{\text{residual vector}}$$

This is an orthogonal decomposition.

$$\underbrace{\begin{bmatrix} -1.2 \\ -0.2 \\ 0.8 \\ -0.2 \\ 0.8 \end{bmatrix}}_{y-\bar{y}} = \underbrace{\begin{bmatrix} -0.7 \\ -0.2 \\ -0.2 \\ 0.3 \\ 0.8 \end{bmatrix}}_{\hat{y}-\bar{y}} + \underbrace{\begin{bmatrix} -0.5 \\ 0 \\ 1 \\ -0.5 \\ 0 \end{bmatrix}}_{\text{residual vector}}$$

Centering by $\bar{y} = 3.2$ keeps it an orthogonal decomposition.

By Pythagorean theorem:

$$\left\| \begin{bmatrix} -1.2 \\ -0.2 \\ 0.8 \\ -0.2 \\ 0.8 \end{bmatrix} \right\|^2 = \left\| \begin{bmatrix} -0.7 \\ -0.2 \\ -0.2 \\ 0.3 \\ 0.8 \end{bmatrix} \right\|^2 + \left\| \begin{bmatrix} -0.5 \\ 0 \\ 1 \\ -0.5 \\ 0 \end{bmatrix} \right\|^2$$

By Pythagorean theorem:

$$\begin{bmatrix} (-1.2)^2+ \\ (-0.2)^2+ \\ (0.8)^2+ \\ (-0.2)^2+ \\ (0.8)^2 \end{bmatrix} = \begin{bmatrix} (-0.7)^2+ \\ (-0.2)^2+ \\ (-0.2)^2+ \\ (0.3)^2+ \\ (0.8)^2 \end{bmatrix} + \begin{bmatrix} -0.5^2+ \\ 0^2+ \\ 1^2+ \\ -0.5^2+ \\ 0^2 \end{bmatrix}$$

By Pythagorean theorem:

$$\underset{\text{Sum of Squares Total}}{[2.8]} = \underset{\text{SS Model}}{[1.3]} + \underset{\text{SS Error}}{[1.5]}$$

By Pythagorean theorem:

$$\underset{\text{SST}}{[2.8]} = \underset{\text{SSM}}{[1.3]} + \underset{\text{SSE}}{[1.5]}$$

By Pythagorean theorem:

$$\underset{\text{SST}}{[2.8]} = \underset{\text{SSM}}{[1.3]} + \underset{\text{SSE}}{[1.5]}$$

SST is the variance of our data's y values (about the mean $\bar{y}$). It is split into two parts:

- SSM: variation in y 's explained by our model.
- SSE: variation in y 's not explained by our model.

By Pythagorean theorem:

$$\underset{\text{SST}}{[2.8]} = \underset{\text{SSM}}{[1.3]} + \underset{\text{SSE}}{[1.5]}$$

The proportion of total variation explained by our model is called the **coefficient of determination**:

$$R = \frac{\text{SSM}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$$

By Pythagorean theorem:

$$\underset{\text{SST}}{[2.8]} = \underset{\text{SSM}}{[1.3]} + \underset{\text{SSE}}{[1.5]}$$

The proportion of total variation explained by our model is called the **coefficient of determination**:

$$R = \frac{1.3}{2.8} = 1 - \frac{1.5}{2.8} = 0.46$$

By Pythagorean theorem:

$$\underset{\text{SST}}{[2.8]} = \underset{\text{SSM}}{[1.3]} + \underset{\text{SSE}}{[1.5]}$$

The proportion of total variation explained by our model is called the **coefficient of determination**:

$$R = \frac{1.3}{2.8} = 1 - \frac{1.5}{2.8} = 0.46$$

"46% of the y's are explained by our linear model relating x's and y's."

By Pythagorean theorem:

$$\underset{\text{SST}}{[2.8]} = \underset{\text{SSM}}{[1.3]} + \underset{\text{SSE}}{[1.5]}$$

Root mean square error (RMSE) for simple linear regression is:

$$\text{RMSE} = \sqrt{\frac{\text{SSE}}{n-2}} = \frac{\|y - \hat{y}\|}{\sqrt{n-2}}$$

Summary:

Model	Fitted Model	Sum of Squares aka Pythagorean Decomposition
$y = ax + b$	$\begin{bmatrix}2\\3\\4\\3\\4\end{bmatrix} = \begin{bmatrix}1&1\\2&1\\2&1\\3&1\\4&1\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} + \begin{bmatrix}e_1\\e_2\\e_3\\e_4\\e_5\end{bmatrix}$	$\underset{y-\bar{y}}{\begin{bmatrix}-1.2\\-0.2\\0.8\\-0.2\\0.8\end{bmatrix}} = \underset{\hat{y}-\bar{y}}{\begin{bmatrix}-0.7\\-0.2\\-0.2\\0.3\\0.8\end{bmatrix}} + \underset{\text{residual vector}}{\begin{bmatrix}-0.5\\0\\1\\-0.5\\0\end{bmatrix}}$

$$R^2 = \frac{\text{SSM}}{\text{SST}} = \frac{\|\hat{y} - \bar{y}\|^2}{\|y - \bar{y}\|^2}$$

FINDING THE LINE OF BEST FIT: IN SAS

```
data points;
    infile datalines dsd;
    input x y;
datalines;
1, 2
2, 3
2, 4
3, 3
4, 4
;
proc reg data=points;
    model y=x / p clm alpha=0.05;
    plot y*x / NOSTAT;
run;
```

Options:

- P = predicted value;
- CLM = confidence interval

The REG Procedure
Model: MODEL1
Dependent Variable: y

Number of Observations Read	5
Number of Observations Used	5

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.30000	1.30000	2.60	0.2053
Error	3	1.50000	0.50000		
Corrected Total	4	2.80000			

Root MSE	0.70711	R-Square	0.4643
Dependent Mean	3.20000	Adj R-Sq	0.2857
Coeff Var	22.09709		

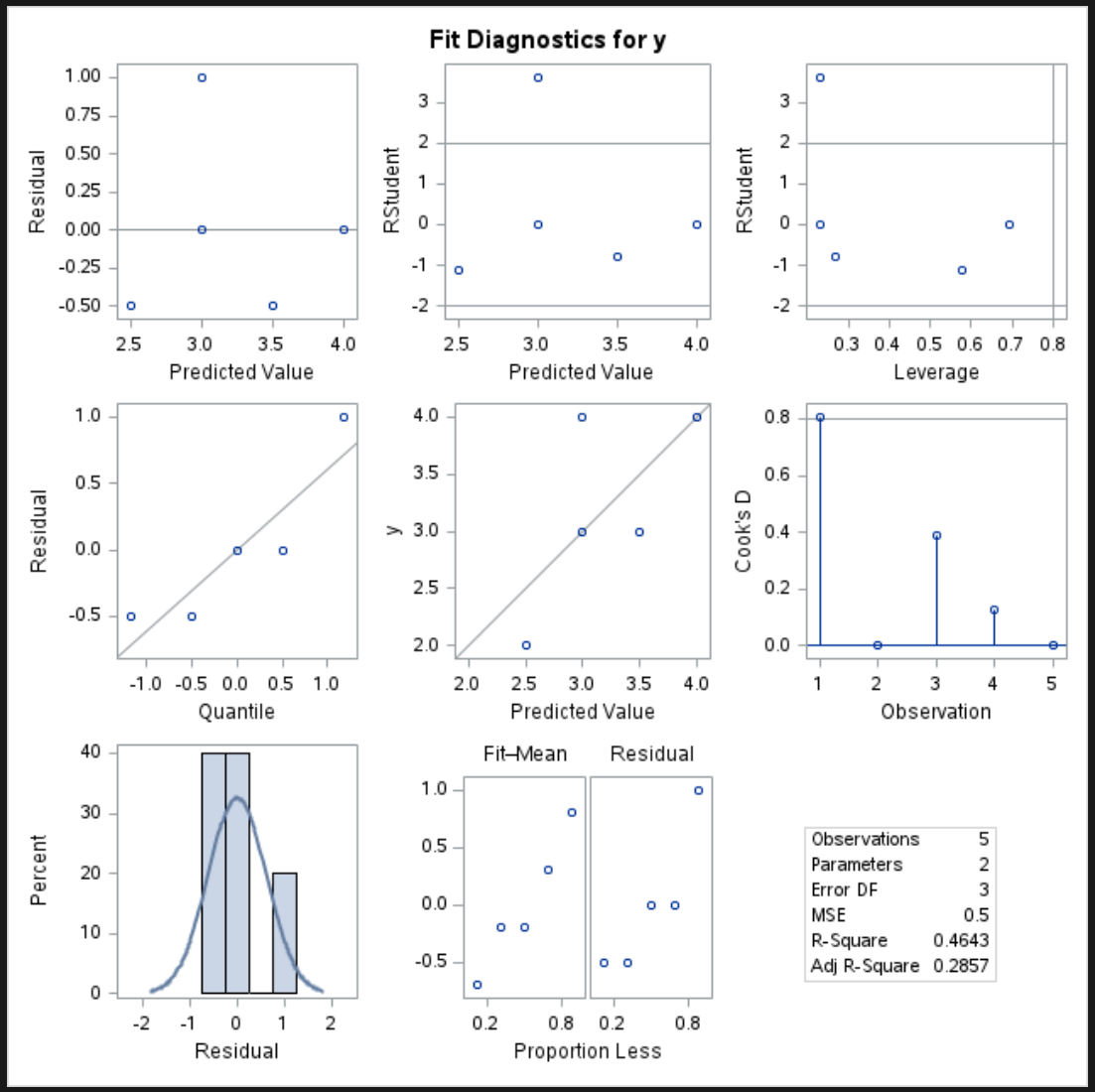
Parameter Estimates

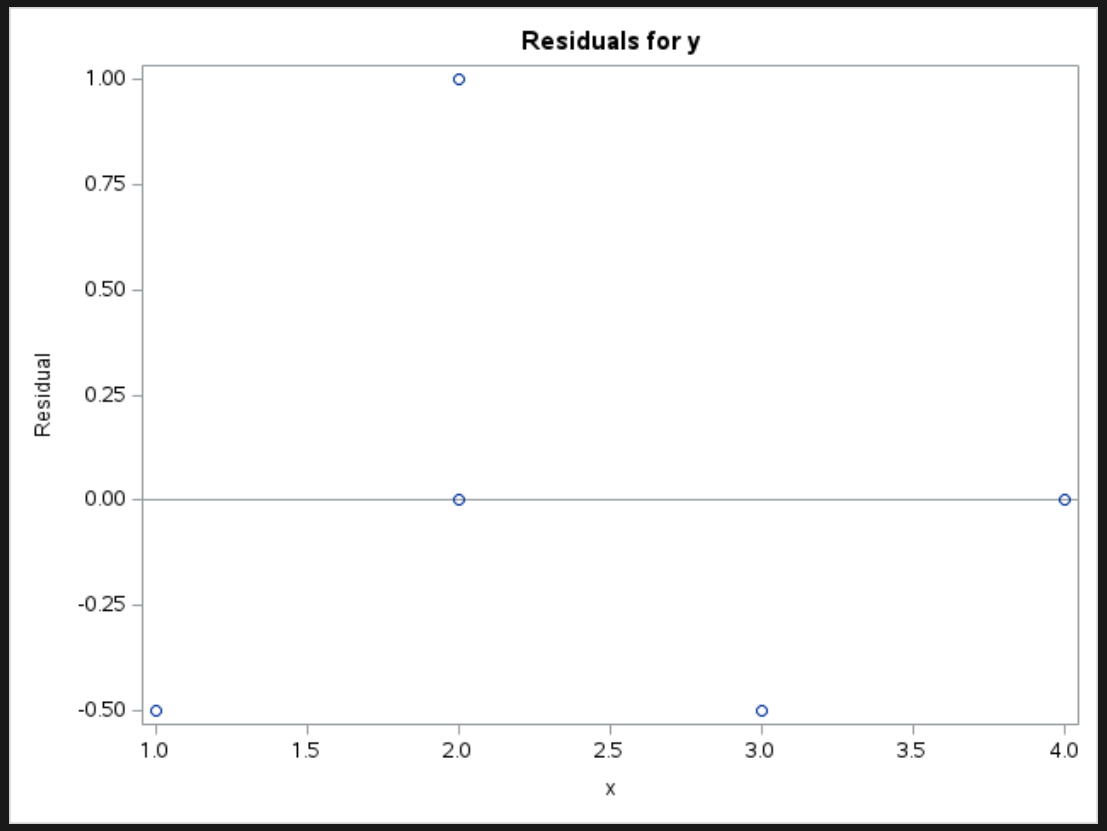
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	2.00000	0.80861	2.47	0.0898
x	1	0.50000	0.31009	1.61	0.2053

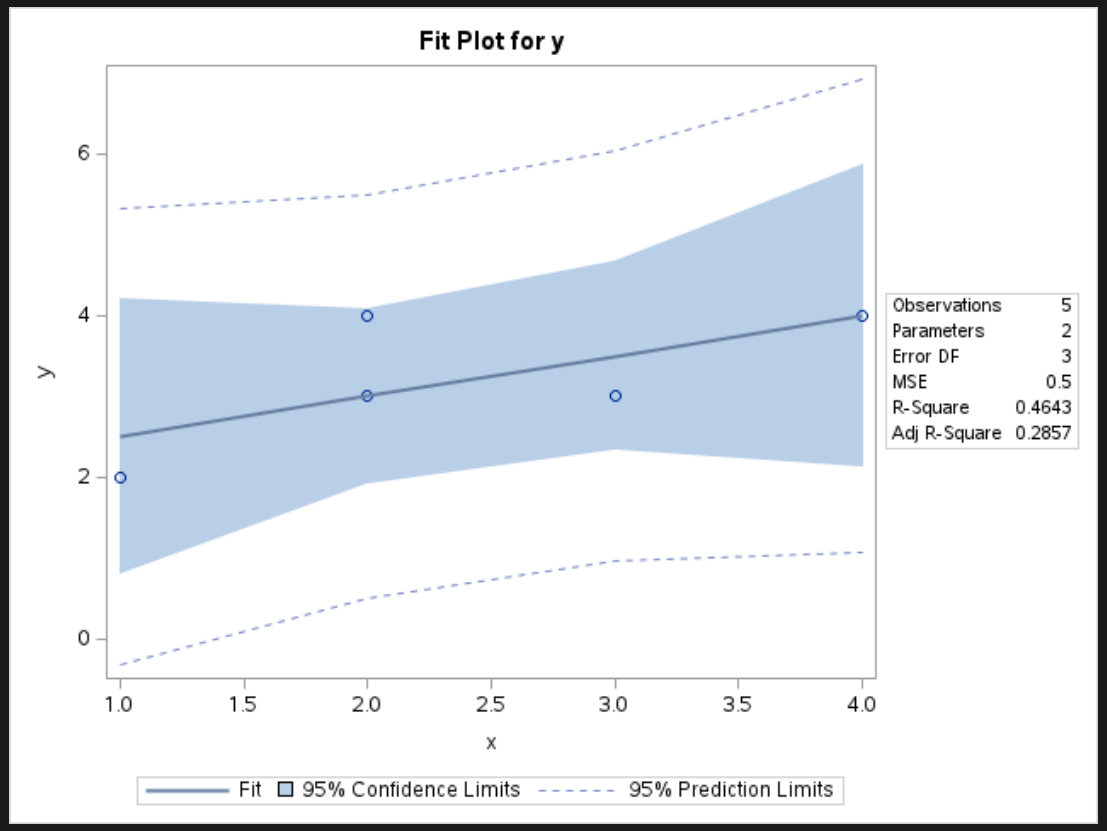
The REG Procedure
Model: MODEL1
Dependent Variable: y

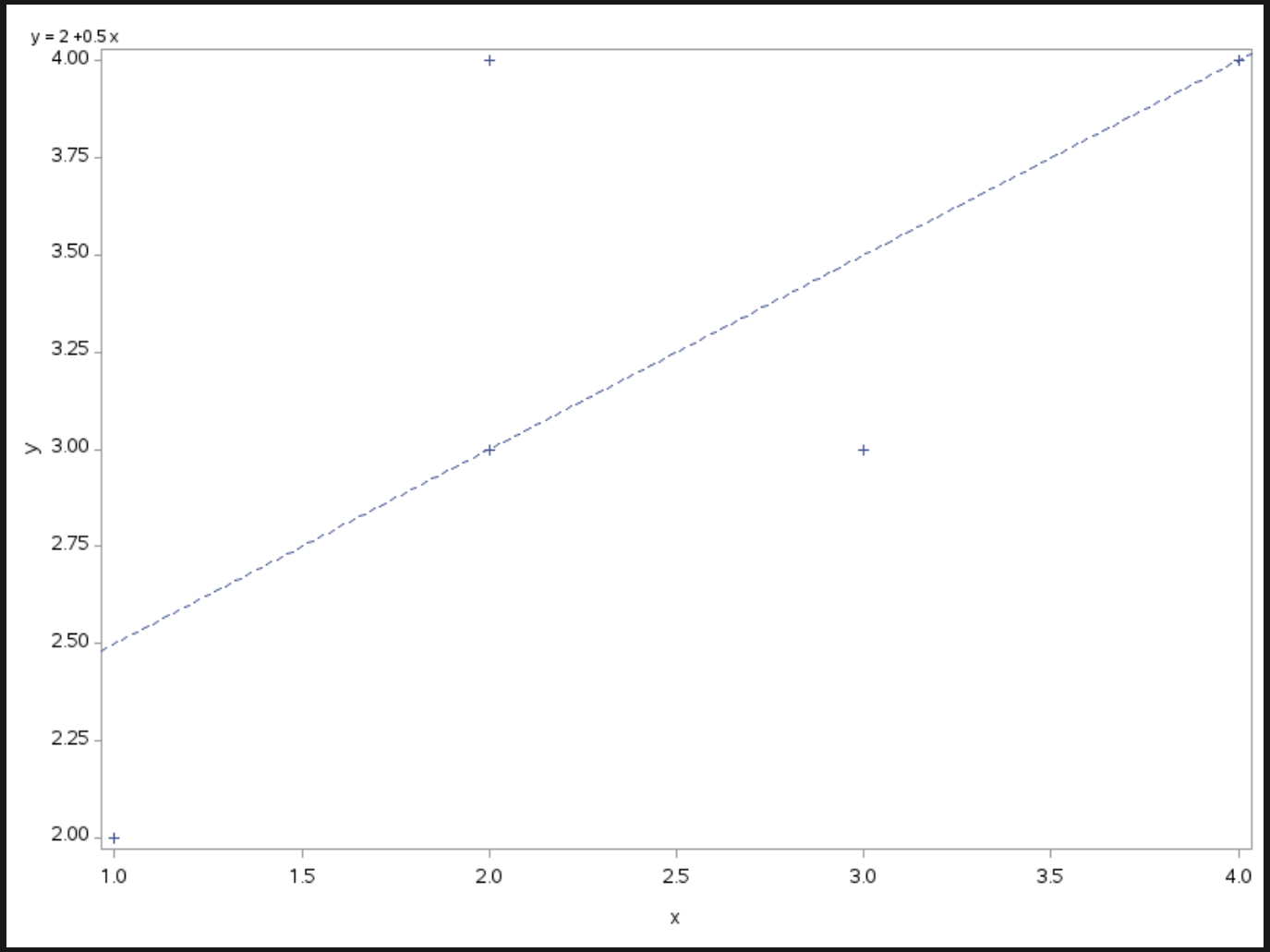
Output Statistics						
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	95% CL Mean		Residual
1	2	2.5000	0.5371	0.7908	4.2092	-0.5000
2	3	3.0000	0.3397	1.9190	4.0810	0
3	4	3.0000	0.3397	1.9190	4.0810	1.0000
4	3	3.5000	0.3669	2.3324	4.6676	-0.5000
5	4	4.0000	0.5883	2.1276	5.8724	0

Sum of Residuals	0
Sum of Squared Residuals	1.50000
Predicted Residual SS (PRESS)	3.55484


Fit Diagnostics for y
Residual
Predicted Value
RStudent
Predicted Value
RStudent
Leverage
Residual
Quantile
y
Predicted Value
Cook's D
Observation
Percent
Residual
Fit–Mean
Residual
Proportion Less
Observations 5
Parameters 2
Error DF 3
MSE 0.5
R-Square 0.4643
Adj R-Square 0.2857


Residuals for y
Residual
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x


Fit Plot for y
y
x
6
4
2
0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Observations 5
Parameters 2
Error DF 3
MSE 0.5
R-Square 0.4643
Adj R-Square 0.2857
Fit
95% Confidence Limits
95% Prediction Limits


y = 2 +0.5 x
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
2.00
y
1.0
1.5
2.0
2.5
3.0
3.5
4.0
x

How good is the fit?

- R^2 closer to 1 than to 0.
- Root mean square error (RMSE) close to 0.
- Residual plot has no systematic pattern, and no changing spread (homoscedasticity) as x varies
- No significant outliers in x- and y- values (which overly influences the line's parameters)

SIMPLE LINEAR REGRESSION: INFERENTIAL

Model $Y = aX + b$ now comes with assumptions on how X, Y behave.

Use this to answer:

- Does X have effect on Y? ($H_0 : a = 0$)
- Predict average response $E[Y|X]$.

The "LINER" assumptions:

- **Linear:**
- **Independent:**
- **Normal:** For any fixed value of x, the response y varies according to a Normal distribution.
- Equal SD (Homoscedasticity): The standard deviation of y is the same for all values of x.
- Random: The data come from a well-designed random sample or randomized experiment.

Assumption 1: Linear. Relationship between variables X and Y is linear: $E[Y|X] = ax + b$

Check by:

- Scatterplot shows a clear linear form.
- Residual plot is scattered about the horizontal line.

Assumption 2: Independence. Individual observations are independent of each other. When sampling without replacement, check the 10% condition.

Check by:

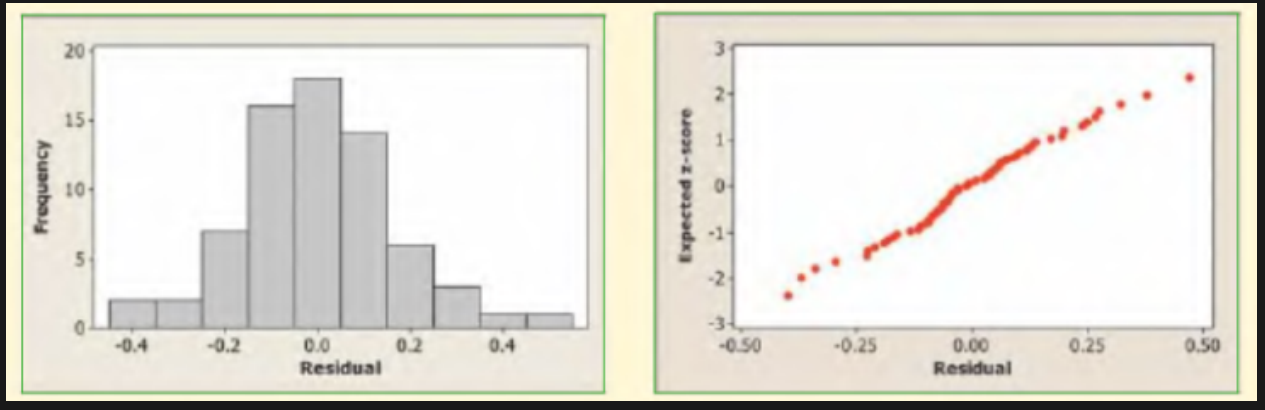
- If we think of x as time, then the next residual should not depend on the previous residual
- If we think of x as time, then there should not be seasonal/cyclical/wavy/sinusoidal pattern in residuals.

Assumption 3: Normal: For any fixed value of x, the response y varies according to a Normal distribution.

Check by:

- Histogram of residuals is single-peaked and somewhat bell-shaped.
- Normal Q-Q plot is very close to linear.

Okay:

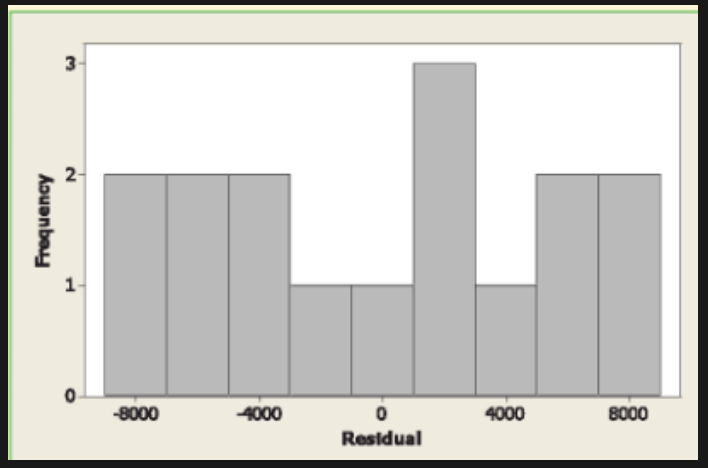


Assumption 3: Normal: For any fixed value of x, the response y varies according to a Normal distribution.

Check by:

- Histogram of residuals is single-peaked and somewhat bell-shaped.
- Normal Q-Q plot is very close to linear.

Okay:

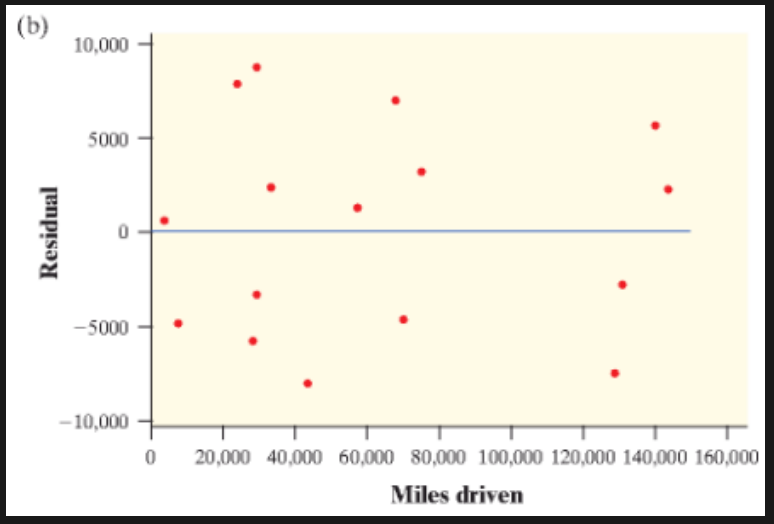


Assumption 4: Equal SD: (Homoscedasticity): The standard deviation of y is the same for all values of x.

Check by:

- Residual plot does not fan out more as you go left or right.

Good:

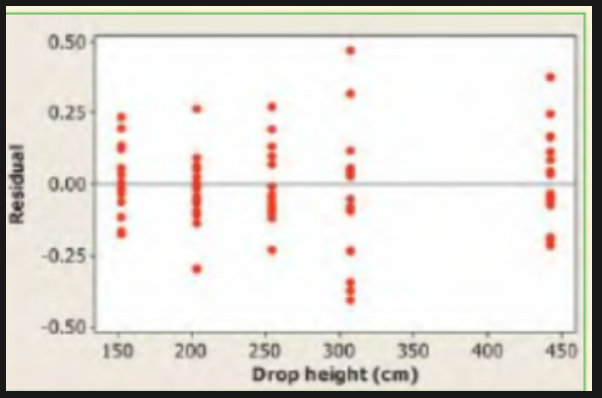


Assumption 4: Equal SD: (Homoscedasticity): The standard deviation of y is the same for all values of x.

Check by:

- Residual plot does not fan out more as you go left or right.

Somewhat okay:

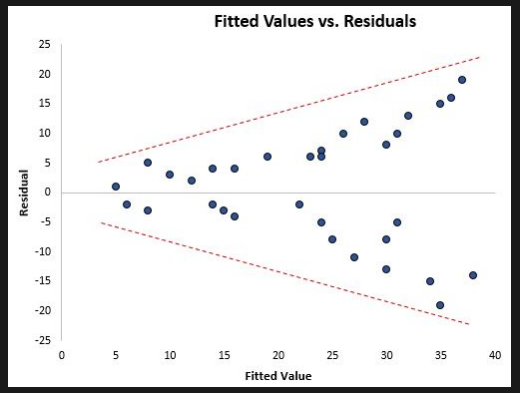


Assumption 4: Equal SD: (Homoscedasticity): The standard deviation of y is the same for all values of x.

Check by:

- Residual plot does not fan out more as you go left or right.

Assumption violated:



Assumption 5: Random: The data come from a well-designed random sample or randomized experiment.

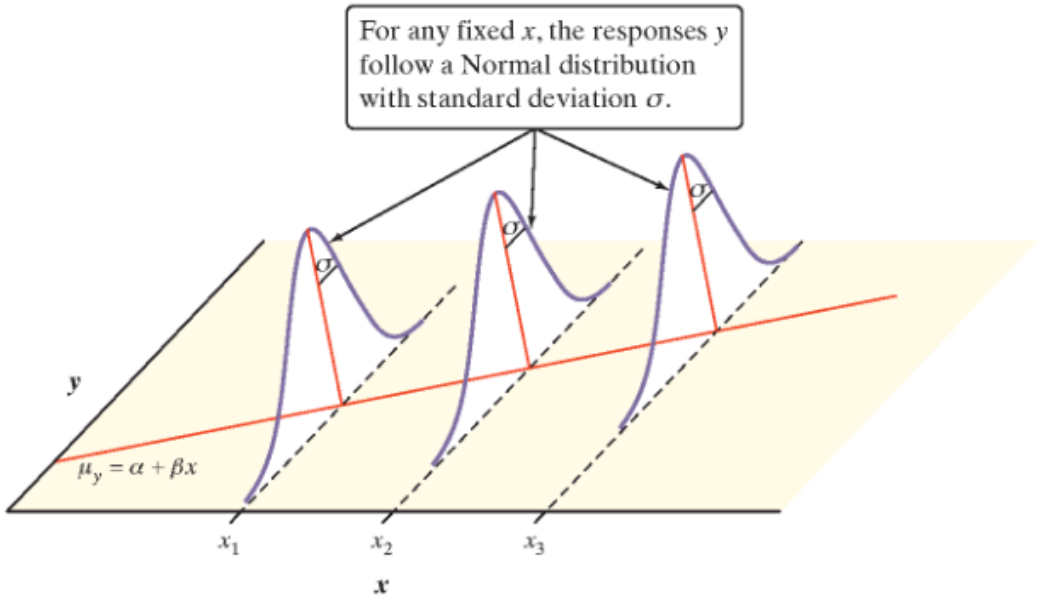


FIGURE 12.6 The regression model when the conditions for inference are met. The line is the population (true) regression line, which shows how the mean response μ_y changes as the explanatory variable x changes. For any fixed value of x, the observed response y varies according to a Normal distribution having mean μ_y and standard deviation σ.

Next time, we:

- continue with confidence interval and hypothesis testing for slope in simple linear regression.
- start multiple linear regression